

x = value of each instalment

Total amount paid in instalments, $A = P \left(1 + \frac{R}{100} \right)^n$

n = number of instalments

EXAMPLE 9. Sapna borrowed some money on compound interest and returned it in 3 yr in equal annual instalments. If rate of interest is 15% per annum and annual instalment is ₹ 486680, then the sum borrowed is.

- a. 111300 b. 122400
c. 1111200 d. 154320

Sol. c. Given that, rate of interest, $R = 15\%$ pa

Annual instalment, $x = ₹ 486680$

Total number of instalments, $n = 3$

$$\therefore P = \left[\frac{x}{\left(1 + \frac{R}{100}\right)} + \frac{x}{\left(1 + \frac{R}{100}\right)^2} + \frac{x}{\left(1 + \frac{R}{100}\right)^3} \right]$$

$$\begin{aligned} \Rightarrow P &= x \left[\frac{100}{(100 + R)} + \frac{(100)^2}{(100 + R)^2} + \frac{(100)^3}{(100 + R)^3} \right] \\ &= 486680 \left[\left(\frac{100}{100 + 15} \right) + \left(\frac{100}{100 + 15} \right)^2 + \left(\frac{100}{100 + 15} \right)^3 \right] \\ &= 486680 \times \left[\frac{20}{23} + \left(\frac{20}{23} \right)^2 + \left(\frac{20}{23} \right)^3 \right] \\ &= 486680 \times \frac{20}{23} \left(1 + \frac{20}{23} + \frac{400}{529} \right) \\ &= 1111200 \\ \therefore \text{Principal borrowed} &= ₹ 1111200 \end{aligned}$$



SOME IMPORTANT RESULTS

1. Simple interest and compound interest are equal for first year period on the same sum and at the same rate.
2. If A and B are the amounts of a certain sum for two consecutive years, then simple interest for 1 yr = $B - A$
3. If certain sum at compound interest becomes x times in n_1 yr and y times in n_2 yr, then $x^{1/n_1} = y^{1/n_2}$

> PRACTICE EXERCISE

- Kiran purchased a scooter for ₹ 24000. The value of scooter is depreciating at the rate of 5% per annum. Then, its value after 3 yr is
(a) ₹ 20577 (b) ₹ 20977 (c) ₹ 20677 (d) ₹ 20877
- If P be the principal amount and the rate of interest be $r\%$ per annum and the compound interest is calculated k times in a year, then what is the amount at the end of n yr?
(a) $P \left(1 + \frac{r}{100k}\right)^{nk}$ (b) $P \left(1 + \frac{kr}{100}\right)^{nk}$
(c) $P \left(1 + \frac{kr}{100}\right)^{n/k}$ (d) $P \left(1 + \frac{kr}{100k}\right)^{n/k}$
- The amount of a certain sum at compound interest for 2 yr at 5% is ₹ 4410. The sum is
(a) ₹ 4000 (b) ₹ 4200 (c) ₹ 3900 (d) ₹ 3800
- A person borrowed ₹ 7500 at 16% compound interest. How much does he have to pay at the end of 2 yr to clear the loan?
(a) ₹ 9900 (b) ₹ 10092 (c) ₹ 11000 (d) ₹ 11052
- If the rate of interest is 10% per annum and is compounded half-yearly, then the principal of ₹ 400 in $3/2$ yr will amount to
(a) ₹ 463.00 (b) ₹ 463.05
(c) ₹ 463.15 (d) ₹ 463.20
- At compound interest, if a certain sum of money doubles in n yr, then the amount will be four fold in
(a) $2n^2$ yr (b) n^2 yr (c) $2n$ yr (d) $4n$ yr
- The simple interest on a certain sum of money for 3 yr at 8% per annum is half the compound interest on ₹ 4000 for 2 yr at 10% per annum. What is the sum placed on simple interest?
(a) ₹ 1550 (b) ₹ 1650 (c) ₹ 1750 (d) ₹ 2000
- What is the least number of complete years in which a sum of money at 20% compound interest will be more than doubled?
(a) 7 (b) 6 (c) 5 (d) 4
- The difference between the simple interest and the compound interest (compounded annually) on ₹ 1250 for 2 yr at 8% per annum will be
(a) ₹ 18 (b) ₹ 2 (c) ₹ 8 (d) ₹ 4
- The compound interest on a sum for 2 yr is ₹ 832 and the simple interest on the same sum at the same rate for the same period is ₹ 800. What is the rate of interest?
(a) 6% (b) 8% (c) 10% (d) 12%
- A saving bank gives interest which compounds annually. Raju deposited ₹ 100 and received ₹ 121 at the end of second year. Rate of compound interest per annum is
(a) 10% (b) 15% (c) 11.5% (d) 20.5%
- The amount of a certain sum at compound interest for 4 yr at 10% is ₹ 4410. The sum is
(a) 3012.08 (b) 3015 (c) 3020.16 (d) 3016.9
- The compound interest on ₹ 5000 for 3 yr at 8% for first year, 10% for second year and 12% for third year will be
(a) ₹ 1560.40 (b) ₹ 1500 (c) ₹ 1565.60 (d) ₹ 1652.80
- An amount of ₹ x at compound interest at 20% per annum for 3 yr becomes y . What is $y : x$?
(a) 3 : 1 (b) 36 : 25
(c) 216 : 125 (d) 125 : 216
- The compound interest on ₹ 2000 for 1 yr at the rate of 8% per annum, when the interest is compounded semi-annually
(a) ₹ 163.20 (b) ₹ 2163.20 (c) ₹ 3153.20 (d) ₹ 1163
- ₹ 16000 invested at 10% per annum compounded semi-annually amounts to ₹ 18522. Then, the period of investment is
(a) $1\frac{1}{2}$ yr (b) 3 yr (c) 2 yr (d) $\frac{5}{2}$ yr
- A sum compounded annually becomes $\frac{25}{16}$ times of itself in 2 yr. Then, the rate of interest per annum is
(a) 25% (b) 20% (c) 15% (d) $7\frac{1}{2}\%$
- A sum of ₹ 3200 invested at 10% per annum compounded quarterly amounts to ₹ 3362, then the time period is
(a) $1\frac{1}{2}$ yr (b) $\frac{1}{2}$ yr (c) 2 yr (d) 1 yr
- A sum amount to ₹ 9680 in 2 yr and to ₹ 10648 in 3 yr compounded annually. Then, the sum and rate of interest, respectively are
(a) ₹ 8000, 10% (b) ₹ 8500, 10%
(c) ₹ 8500, 9% (d) ₹ 8000, 9%
- If the value of a machine depreciates by 10% of its value at the beginning of the year and its present value is estimated as ₹ 10935, then what was its value three years back?
(a) ₹ 15000 (b) ₹ 7000
(c) ₹ 8050 (d) None of these

HINTS AND SOLUTIONS

1. (a) Given, $P = ₹ 24000$, $R = 5\%$ per annum and $n = 3$ yr

$$\begin{aligned}\therefore A &= P \left(1 + \frac{R}{100}\right)^n \quad [\text{Rule 8 (i)}] \\ &= 24000 \left(1 + \frac{5}{100}\right)^3 \\ &= 24000 \times \left(\frac{95}{100}\right)^3 \\ &= ₹ 20577\end{aligned}$$

2. (a) Given, principal amount = ₹ P
Rate of interest, $R = \frac{r}{k}\%$ per annum and
Time, $T = nk$

$$\therefore A = P \left(1 + \frac{R}{100}\right)^T \quad [\text{Rule 1. (i)}]$$

$$\therefore A = P \left(1 + \frac{r}{100k}\right)^{nk}$$

3. (a) Let the principal be = ₹ x .
Given, $n = 2$ yr, $R = 5\%$, $A = ₹ 4410$

$$\therefore A = P \left(1 + \frac{R}{100}\right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore 4410 = x \left(1 + \frac{5}{100}\right)^2$$

$$\Rightarrow 4410 = x \left(\frac{21}{20}\right)^2$$

$$\Rightarrow x = \frac{4410 \times 400}{441} = ₹ 4000$$

4. (b) Given, $P = ₹ 7500$, $R = 16\%$ and
 $n = 2$ yr

$$\therefore A = P \left(1 + \frac{R}{100}\right)^n \quad [\text{Rule 1. (i)}]$$

$$= 7500 \left(1 + \frac{16}{100}\right)^2$$

$$= 7500 \left(\frac{116}{100}\right)^2$$

$$= 7500 \times \frac{29}{25} \times \frac{29}{25}$$

$$= 12 \times 841 = ₹ 10092$$

5. (b) Refer to example 2.

6. (c) Let the principal be ₹ x .

$\therefore$ Rate = ₹ R , Amount = ₹ $2x$
and Time = n yr

$$\therefore A = P \left(1 + \frac{R}{100}\right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore x \left(1 + \frac{R}{100}\right)^n = 2x$$

$$\Rightarrow \left(1 + \frac{R}{100}\right)^n = 2 \quad \dots(i)$$

Let it becomes four fold in N yr.

$$\text{Then, } x \left(1 + \frac{R}{100}\right)^N = 4x$$

$$\Rightarrow \left(1 + \frac{R}{100}\right)^N = 4$$

$$\Rightarrow 2^2 = \left(1 + \frac{R}{100}\right)^N$$

$$\Rightarrow \left(1 + \frac{R}{100}\right)^{2n} = \left(1 + \frac{R}{100}\right)^N \quad [\text{from Eq. (i)}]$$

$$\therefore N = 2n \text{ yr}$$

7. (c) Let the principal amount be ₹ P .

By given condition, $SI = \frac{1}{2} CI$

$$\Rightarrow \frac{P \times 8 \times 3}{100} = \frac{1}{2} \left[4000 \left(1 + \frac{10}{100}\right)^2 - 4000 \right]$$

[Rule 1 (ii)]

$$\Rightarrow \frac{24P}{100} = \frac{1}{2} \left[4000 \times \frac{121}{100} - 4000 \right]$$

$$= \frac{1}{2} [4840 - 4000]$$

$$\Rightarrow \frac{24P}{100} = 420$$

$$\therefore P = \frac{420 \times 100}{24} = ₹ 1750$$

8. (d) Let the sum of money be ₹ P

$\therefore$ Amount = ₹ $2P$

$$\Rightarrow A = P \left(1 + \frac{R}{100}\right)^n \quad [\text{Rule 1. (i)}]$$

$$\Rightarrow 2P = P \left(1 + \frac{20}{100}\right)^n$$

$$\Rightarrow \frac{2P}{P} = \left(\frac{6}{5}\right)^n$$

$$\Rightarrow 2 = \left(\frac{6}{5}\right)^n$$

On putting $n = 4$, we get

$$\left(\frac{6}{5}\right)^4 = \frac{1296}{625} = 2 \text{ (approx)}$$

$\therefore$ Least number of years = 4

9. (c) Given, $P = ₹ 1250$, $R = 8\%$ per annum and $n = 2$ yr

$$\therefore \text{Simple interest} = \frac{P \times R \times T}{100}$$

$$\therefore \text{Simple interest} = \frac{1250 \times 8 \times 2}{100}$$

$$= ₹ 200$$

$\therefore$ Compound interest

$$= P \left[\left(1 + \frac{R}{100}\right)^n - 1 \right] \quad [\text{Rule 1. (ii)}]$$

$$\therefore CI = 1250 \left(1 + \frac{8}{100}\right)^2 - 1250$$

$$= 1250 \times \left(\frac{108}{100}\right)^2 - 1250$$

$$= 1458 - 1250 = ₹ 208$$

$\therefore$ Difference in SI and CI

$$= 208 - 200 = ₹ 8$$

Shortcut Method

Here, $p = ₹ 1250$ and $R = 8\%$

$$\therefore \text{Required difference} = \frac{PR^2}{(100)^2} \quad [\text{Rule 6. (i)}]$$

$$= \frac{1250 \times 64}{100 \times 100} = ₹ 8$$

10. (b) Given, $CI = ₹ 832$, $SI = ₹ 800$,
 $n = 2$ yr

$$CI = P \left[\left(1 + \frac{R}{100}\right)^n - 1 \right]$$

[Rule 1. (ii)]

$$\therefore 832 = P \left[\left(1 + \frac{R}{100}\right)^2 - 1 \right] \quad \dots(i)$$

$$\text{Also, } SI = \frac{P \times R \times T}{100}$$

$$\Rightarrow 800 = \frac{P \times R \times 2}{100}$$

$$\Rightarrow P = \frac{40000}{R}$$

From Eq. (i),

$$832 = \frac{40000}{R} \left(\frac{R^2}{10000} + \frac{2R}{100} \right)$$

$$\Rightarrow 832 = \frac{40000}{100} \left(\frac{R}{100} + 2 \right)$$

$$\Rightarrow 832 = 4R + 800$$

$$\therefore R = \frac{32}{4} = 8\%$$

- 11. (a)** Given, principal (P) = ₹ 100
Amount (A) received after 2 yr = ₹ 121
Let rate of interest = $R\%$ per annum

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore 121 = 100 \left(1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \frac{121}{100} = \left(\frac{100 + R}{100} \right)^2$$

$$\Rightarrow \left(\frac{11}{10} \right)^2 = \left(\frac{100 + R}{100} \right)^2$$

$$\Rightarrow \frac{11}{10} = \frac{100 + R}{100}$$

$$\Rightarrow 100 + R = \frac{11 \times 100}{10}$$

$$\Rightarrow 100 + R = 110$$

$$\therefore R = 110 - 100 = 10$$

$$\therefore \text{Rate of interest} = 10\%$$

- 12. (a)** Refer to example 3.

- 13. (d)** Given, $P = ₹ 5000$,

$$R_1 = 8\%, R_2 = 10\%, R_3 = 12\%$$

$$\text{and } n_1 = n_2 = n_3 = 1 \text{ yr}$$

$$\therefore \text{Amount}$$

$$= P \left(1 + \frac{R_1}{100} \right)^{n_1} \left(1 + \frac{R_2}{100} \right)^{n_2} \left(1 + \frac{R_3}{100} \right)^{n_3}$$

$$[\text{Rule 5}]$$

$$= \left[5000 \times \left(1 + \frac{8}{100} \right) \left(1 + \frac{10}{100} \right) \left(1 + \frac{12}{100} \right) \right]$$

$$= \left(5000 \times \frac{27}{25} \times \frac{11}{10} \times \frac{28}{25} \right) = ₹ 6652.80$$

$$\therefore \text{Compound interest} = 6652.80 - 5000$$

$$= ₹ 1652.80$$

- 14. (c)** Given, $P = ₹ x$, $R = 20\%$ per annum,
 $n = 3$ yr

$$\text{and } A = ₹ y$$

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore y = x \left(1 + \frac{20}{100} \right)^3$$

$$\Rightarrow y = x \left(\frac{6}{5} \right)^3 \Rightarrow \frac{y}{x} = \frac{216}{125}$$

$$\Rightarrow y : x = 216 : 125$$

- 15. (a)** Given, principal (P) = ₹ 2000

$$\text{Time} = 1 \text{ yr}$$

$$\text{Rate of interest} = 8\% \text{ per annum}$$

$$\text{Since, the interest is compounded semi-annually.}$$

$$\text{Then, rate } (R) = \frac{8}{2}\% = 4 \text{ half-yearly}$$

$$\text{and time } (n) = 1 \times 2 = 2 \text{ half-yearly}$$

$$\therefore \text{CI} = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right]$$

$$[\text{Rule 1. (ii)}]$$

$$\text{CI} = 2000 \left(1 + \frac{4}{100} \right)^2 - 2000$$

$$= 2000 \times \left(\frac{26}{25} \right)^2 - 2000$$

$$= 2163.20 - 2000 = ₹ 163.20$$

- 16. (a)** Given, principal (P) = ₹ 16000

$$\text{Amount received at the end of period } (A) = ₹ 18522$$

$$\text{Let time} = t \text{ yr}$$

$$\therefore n = 2t$$

$$\text{Rate} = 10\% \text{ per annum i.e } 5\% \text{ half-yearly}$$

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore 18522 = 16000 \left(1 + \frac{5}{100} \right)^n$$

$$\Rightarrow \left(\frac{18522}{16000} \right) = \left(\frac{21}{20} \right)^n$$

$$\Rightarrow \left(\frac{9261}{8000} \right) = \left(\frac{21}{20} \right)^n \Rightarrow \left(\frac{21}{20} \right)^3 = \left(\frac{21}{20} \right)^n$$

$$\Rightarrow n = 3$$

$$\therefore n = 2t \Rightarrow 2t = 3$$

$$\therefore t = \frac{3}{2} \text{ yr} = 1 \frac{1}{2} \text{ yr}$$

- 17. (a)** Let sum/ principal be ₹ x and the rate be $R\%$ per annum.

$$\therefore \text{Amount, } A = \frac{25}{16} x \text{ and } n = 2 \text{ yr}$$

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore \frac{25}{16} x = x \left(1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \frac{25}{16} = \left(1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \left(\frac{5}{4} \right)^2 = \left(1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \frac{5}{4} = \left(1 + \frac{R}{100} \right) \Rightarrow \frac{R}{100} = \frac{1}{4}$$

$$\therefore R = 25\%$$

- 18. (b)** Here, $P = ₹ 3200$,

$$A = ₹ 3362$$

$$[\text{since, amount is payable quarterly}]$$

$$\therefore R = 10\% \text{ per annum} = \frac{10}{4}\% \text{ quarterly}$$

$$\text{and let time} = t$$

$$\Rightarrow n = 4t$$

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\Rightarrow 3362 = 3200 \left(1 + \frac{10}{4 \times 100} \right)^n$$

$$\Rightarrow \frac{3362}{3200} = \left(\frac{410}{400} \right)^n \Rightarrow \left(\frac{41}{40} \right)^2 = \left(\frac{41}{40} \right)^n$$

$$\Rightarrow n = 2 \Rightarrow 4t = 2 \text{ yr} \quad [\text{as } n = 4t]$$

$$\therefore t = \frac{1}{2} \text{ yr}$$

- 19. (a)** Given, $A_1 = ₹ 9680$, $A_2 = ₹ 10648$
and $n = 2$ yr

$$\therefore \text{Rate of compound interest}$$

$$= \frac{A_2 - A_1}{A_1} \times 100\% \quad [\text{Rule 7 (i)}]$$

$$= \frac{10648 - 9680}{9680} \times 100\% = 10\%$$

$$\therefore \text{Sum} = A_1 \left(\frac{A_2}{A_1} \right)^n \quad [\text{Rule 7 (ii)}]$$

$$= 9680 \left(\frac{10648}{9680} \right)^2$$

$$= 9680 \left(\frac{110}{121} \right)^2 = 9680 \left(\frac{10}{11} \right)^2$$

$$= \frac{9680 \times 100}{121} = ₹ 8000$$

- 20. (a)** Let the value of machine 3 yr ago be ₹ x .

$$\text{and given, } P = ₹ 10935, R = 10\% \text{ and } n = 3 \text{ yr}$$

$$\therefore x = P \left(1 - \frac{R}{100} \right)^n \quad [\text{Rule 8 (i)}]$$

$$\therefore x \left(1 - \frac{10}{100} \right)^3 = 10935$$

$$\Rightarrow x \left(\frac{90}{100} \right)^3 = 10935$$

$$\therefore x = \frac{10935 \times 10 \times 10 \times 10}{9 \times 9 \times 9} = ₹ 15000$$

- 21. (d)** Let the sum be ₹ x and rate be $R\%$ per annum.

$$\therefore \text{Amount, } A = ₹ 2x$$

$$\text{and time, } n = 15 \text{ yr}$$

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\text{Then, } x \left(1 + \frac{R}{100} \right)^{15} = 2x$$

$$\Rightarrow \left(1 + \frac{R}{100} \right)^{15} = 2 \quad \dots(i)$$

$$\text{Suppose the sum becomes eight times in 'n' yr, then}$$

$$x \left(1 + \frac{R}{100} \right)^n = 8x \quad [\text{by given condition}]$$

$$\Rightarrow \left(1 + \frac{R}{100} \right)^n = 8 = 2^3 \quad \dots(ii)$$